

FURTHER MATHEMATICS

Unit 3

Targeted Evaluation Task for School-assessed Coursework 1



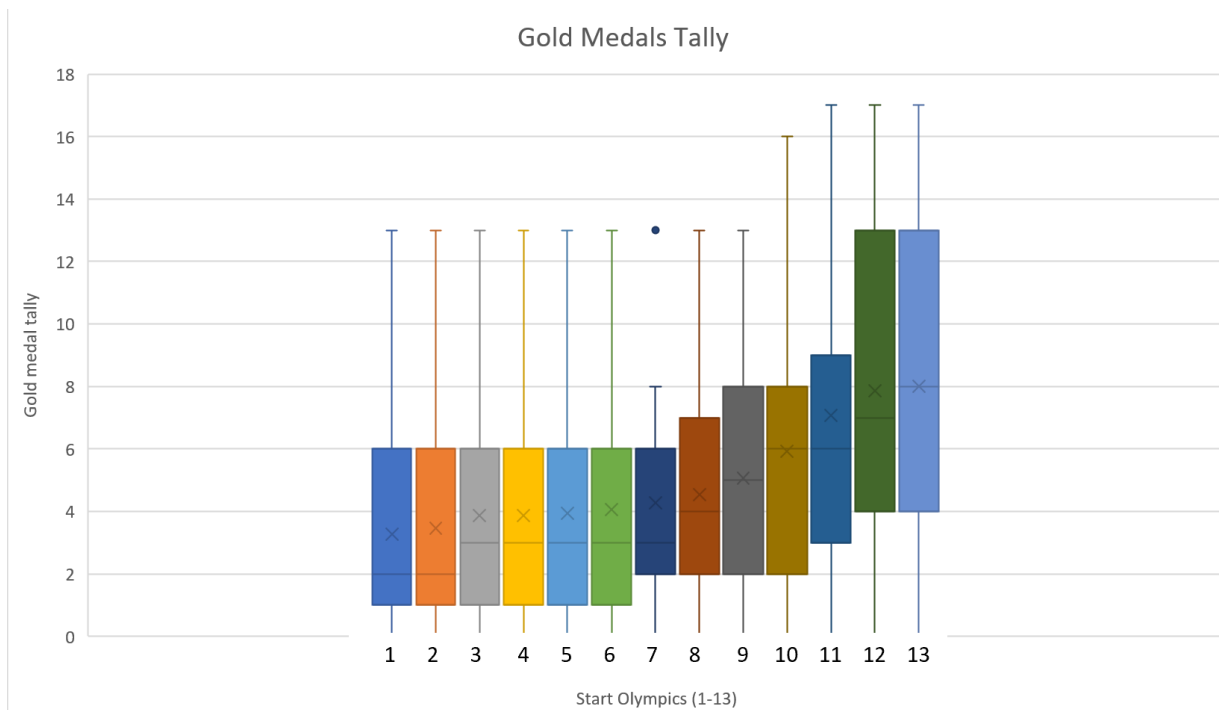
2020 Application Task on Data Analysis for Outcomes 1, 2 & 3

SOLUTIONS & RESPONSE GUIDE

Question 1

- a. Solutions will vary depending on the Olympiad chosen, see the table below ordered by the starting Olympiad. Students should select an appropriate graphical display such as a histogram or stem and leaf plot and comment on the shape produced.

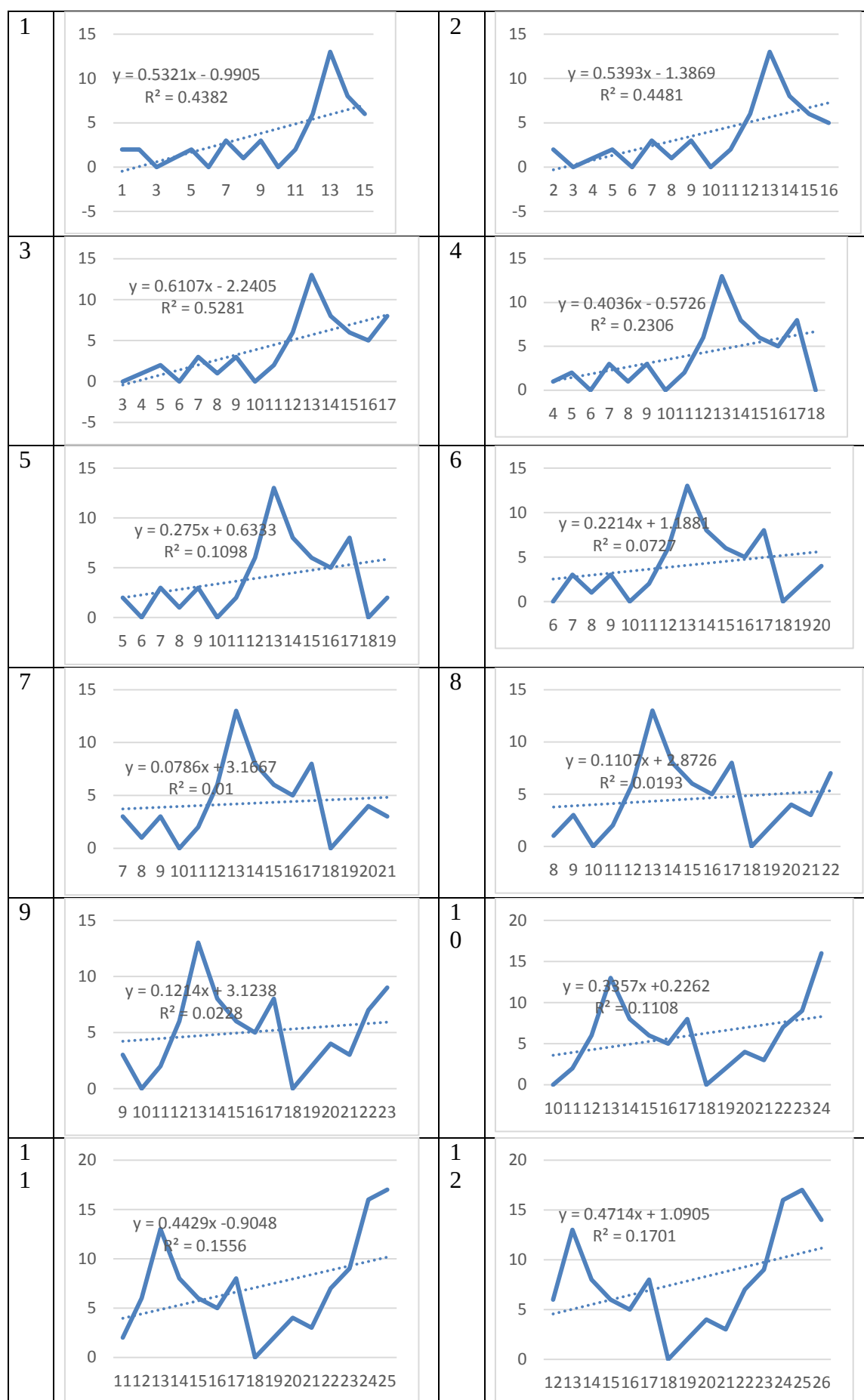
Starting Olympiad	Mean	Median	Standard Deviation	Shape
1	3.27	2	3.59	+ skew
2	3.47	2	3.60	+ skew
3	3.87	3	3.76	+ skew
4	3.87	3	3.76	+ skew
5	3.93	3	3.71	+ skew
6	4.07	3	3.67	+ skew
7	4.27	3	3.51	+ skew with outlier
8	4.53	4	3.56	+ skew
9	5.07	5	3.59	+ skew
10	5.93	6	4.51	+ skew
11	7.07	6	5.02	+ skew
12	7.87	7	5.11	+ skew
13	8.00	8	5.08	Symmetrical

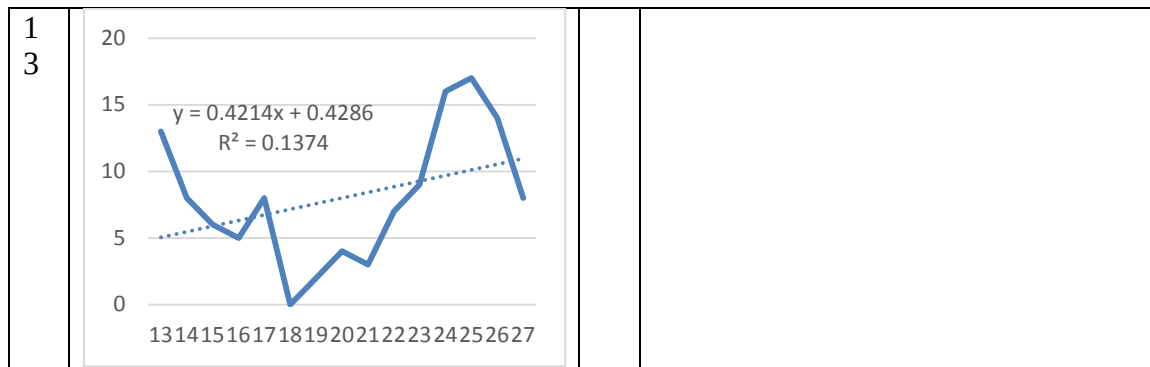


O1 – 2 marks, O2 – 2 marks

- b. Solutions will vary depending on the Olympiad chosen, see the table below ordered by the starting Olympiad, using Olympiad as the x-axis

2020 FURTHER MATHEMATICS SCHOOL-ASSESSED COURSEWORK





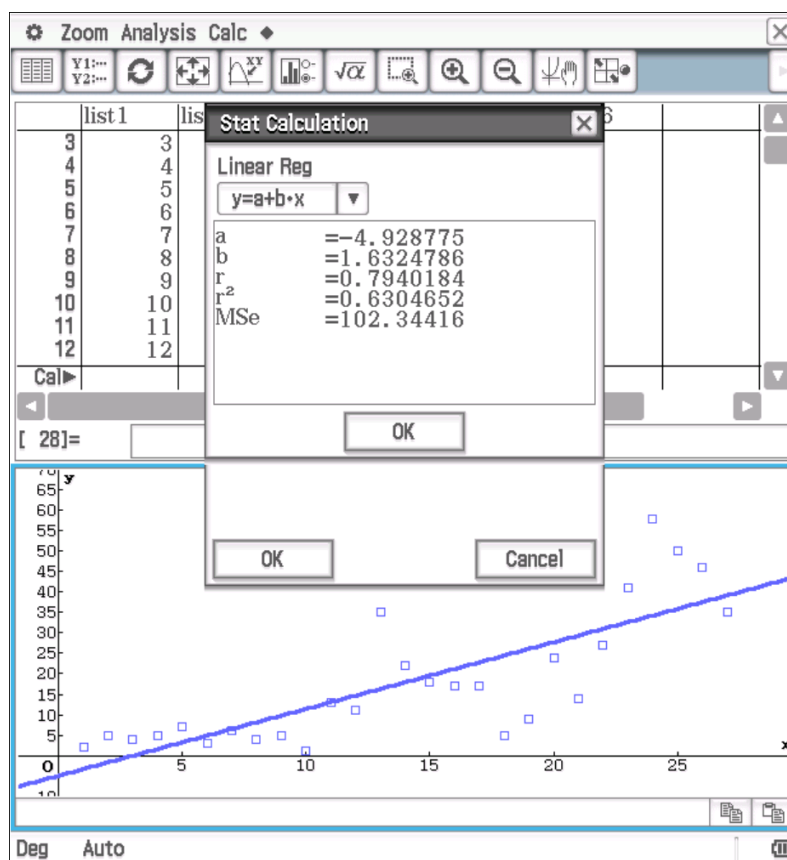
O1 – 3 marks

- c. Solutions will vary depending on the starting Olympiad chosen. See the regression line and r value of the regression line to identify the strength and trend of the number of gold medals. Students should refer to the peak at the Melbourne Olympics, decrease to the Montreal Olympics, steady increase to the Athens Olympics and/or the decline following Sydney.

O2 – 3 marks

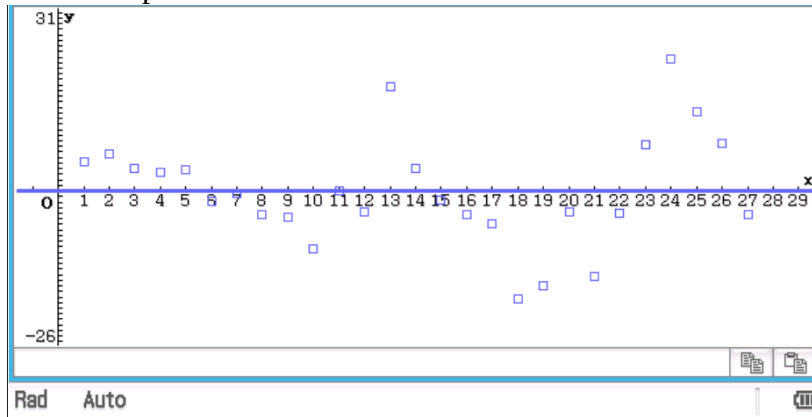
Question 2

- a. The equation of the regression line is $total\ medals = -4.929 + 1.632 \times Olympiad$. $r=0.794$, so there is a relatively strong positive correlation between the number of medals and the Olympiad. Students should refer to the peak at the Melbourne Olympics, decrease to the Montreal Olympics, steady increase to the Sydney Olympics and the decline following Sydney.



O1 – 1 mark, O2 – 5 marks, O3 – 2 marks

- b. Excluding the outliers around the Melbourne Olympics there is a trend in the residual to go from positive to negative and back to positive, which suggests that, although the linear relationship shows a strong correlation, there may be a better non-linear relationship.



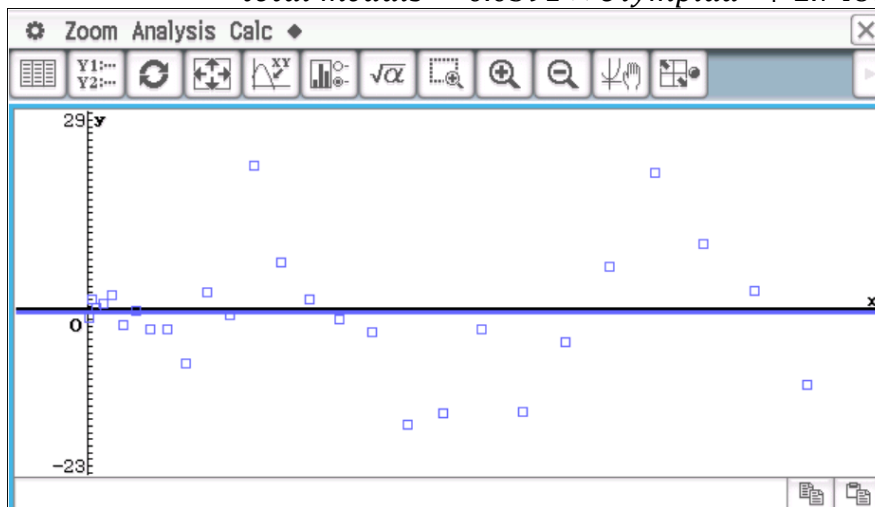
O2 – 1 mark, O3 – 2 marks

- c. The shape of the scatterplot suggests an Olympiad squared transformation.

$r = 0.8298$, $r^2 = 0.6885$, pattern of residual plot less defined.

A suitable model for the number of gold medals is:

$$\text{total medals} = 0.0591 \times \text{Olympiad}^2 + 2.7489$$



O2 – 2 marks, O3 – 2 marks

d. *total medals in London* $= 0.0591 \times 27^2 + 2.7489$
 $= 45.83 = 46$ medals
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Actual results: 35 medals

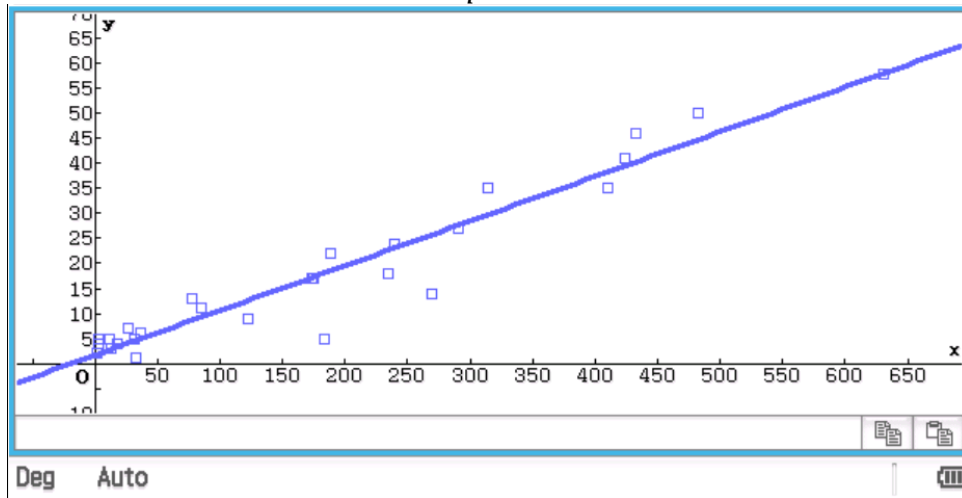
total medals in Rio de Janeiro $= 0.0591 \times 28^2 + 2.7489$
 $= 49.083 = 49$ medals
Actual results 29 medals

The predicted number of medals won in London was more accurate than the calculation for Rio de Janeiro. The prediction for the London games is interpolation, whereas, the prediction for Rio de Janeiro is extrapolation. Also there has been a downward trend in the number of medals won from the 2000 Sydney Olympics to the 2016 Rio de Janeiro Olympics, which is not reflected in the calculated model.

O2 – 2 mark, O3 – 1 marks

Question 3

a. $\text{medals won} = 0.0893 \times \text{competitors} + 1.6873$



O1 – 2 marks, O3 – 2 marks

b. $r = 0.9613, r^2 = 0.9242$

92.4% of the variation in the number of medals won by Australia at the Olympics can be explained by the number of competitors that Australia sends.

O2 – 1 mark, O3 – 1 mark

c. London Olympics = $0.0893 \times 410 + 1.6873 = 38.30 = 38 \text{ medals}$

Rio de Janeiro = $0.0893 \times 422 + 1.6873 = 39.37 = 39 \text{ medals}$

O1 – 2 marks

- d. Australia won 35 medals at the London Olympics, which was close to the predicted 38 medals (within 10% - interpolation), however the prediction for the Rio de Janeiro Olympics of 39 was much higher than the actual result of 29 (extrapolation).

O2 – 2 marks

- e. The prediction for the number of medals won using the number of competitors Australia sends has a greater accuracy than the prediction using the Olympiad. The percentage error of the predictions is much lower using the number of competitors sent and the residual plot shows less patterns than using the Olympiad.

O2 – 2 marks